

Early Prediction of Pediatric Sepsis Using an Explainable CatBoost Model on the PIC Database:

A SHAP-Based Feature Importance Analysis — Literature Review

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1. Introduction and Clinical Background

Sepsis is one of the most life-threatening conditions encountered in pediatric intensive care units (PICUs), defined as a dysregulated host response to infection resulting in organ dysfunction. Despite advances in critical care, pediatric sepsis continues to carry significant morbidity and mortality worldwide. Early identification and intervention are widely recognized as the primary determinants of patient outcomes; however, timely recognition remains a clinical challenge due to the non-specific presentation of sepsis, particularly in younger children whose physiological responses differ substantially from adults (Sanchez-Pinto et al., 2024).

Traditional scoring systems such as Pediatric Logistic Organ Dysfunction-2 (PELOD-2) and Systemic Inflammatory Response Syndrome (SIRS) criteria have served as standard tools for sepsis detection in PICUs. However, these tools lack the specificity required for reliable early prediction — SIRS criteria, in particular, identify inflammatory responses from both infectious and non-infectious causes, reducing their clinical discriminatory value (Le et al., 2019). This limitation has created a strong impetus for data-driven, machine learning (ML)-based approaches that leverage the richness of electronic health records (EHR) available in modern ICU settings.

A further challenge in this domain is the under-representation of pediatric populations in the ML literature. Most existing sepsis prediction models have been developed using adult ICU databases, raising serious concerns about their generalizability to children. Additionally, the majority of published work originates from Western healthcare systems, limiting applicability to pediatric populations in other clinical and demographic contexts, including China (Tennant et al., 2024). This literature review examines the current state of ML-based sepsis prediction, with specific focus on pediatric applications, the PIC database, and the role of explainable AI (XAI) methods in improving clinical utility.

2. Machine Learning Approaches to Sepsis Prediction

The application of machine learning to sepsis prediction has expanded substantially over the past decade. Algorithms trained on vital signs, laboratory values, medications, and demographic data have demonstrated high predictive accuracy — often exceeding that of conventional clinical scores. A 2023 systematic review of 42 studies confirmed the propensity of ML models for early sepsis prediction, with AUROC values commonly exceeding 0.85 across diverse ICU settings (Yang et al., 2023). However, that same review documented a critical limitation: the overwhelming majority of studies relied on MIMIC-II or MIMIC-III — adult ICU databases — with pediatric cohorts largely excluded.

In the pediatric domain specifically, Le et al. (2019) demonstrated that a machine learning algorithm could predict severe sepsis onset 4 hours before clinical recognition in children aged 2–17, achieving an AUROC of 0.916 and significantly outperforming both PELOD-2 and SIRS criteria on a cohort of 9,486 pediatric patients. This foundational study established the feasibility of ML-based early warning for pediatric sepsis, but relied on a private institutional dataset from the University of California, San Francisco — limiting reproducibility and broader validation.

Gradient boosting ensemble methods have consistently demonstrated competitive or superior performance in clinical prediction tasks involving tabular EHR data. Among these, CatBoost has emerged as a strong performer due to its native handling of categorical features, robustness to overfitting on imbalanced datasets, and reduced need for extensive hyperparameter tuning. A 2025 study trained multiple ML models — including CatBoost — on over 63,875 PICU encounters using the Phoenix international pediatric sepsis criteria, achieving an AUROC of 0.98 using readily available EMR variables (Chanci et al., 2025). While this represents the strongest reported performance for CatBoost in pediatric sepsis prediction to date, the study was conducted on a private US institutional dataset, and no SHAP-based explainability analysis was integrated into the primary methodology.

Han et al. (2024) applied CatBoost alongside extreme gradient boosting for the detection of bloodstream infections in critically ill children, using 1,806 measurements and standard sepsis criteria, and found CatBoost competitive with XGBoost in this adjacent clinical task. Shi et al. (2025) applied XGBoost and a GRU-based recurrent neural network to pediatric sepsis prediction at Guangzhou Women and Children's Medical Center in China, achieving an AUROC of 0.915 with a median lead time of 6.2 hours before clinical diagnosis. While this study provides the most directly comparable Chinese pediatric data, it relied on a private institutional dataset rather than a publicly available database, limiting reproducibility.

3. The PIC Database and Pediatric ICU Research

The Paediatric Intensive Care (PIC) database represents the most significant publicly available resource for pediatric-specific ML research in a Chinese clinical context. Developed by Zeng et al. (2020), PIC is a large, single-centre, bilingual database comprising de-identified health records of critically ill children admitted to a major children's hospital in China. It includes vital signs, medications, laboratory measurements, fluid balance, ICD-coded diagnoses, microbiology results, and survival data — structured analogously to MIMIC but specific to pediatric patients with over 12,881 records.

Despite its availability and clinical richness, published ML research using the PIC database remains limited in both volume and scope. Critically, no published study has applied ML to pediatric sepsis onset prediction as the primary clinical target using this database. Existing PIC-based ML studies have examined adjacent clinical questions: Shen et al. (2025) applied CatBoost with SHAP to predict 28-day mortality in pediatric acute kidney injury (AKI) patients using 3,624

PIC records, identifying lactate as a key predictor. Zhou et al. (2025) developed an interpretable ML model using SHAP for early prediction of disseminated intravascular coagulation (DIC) in critically ill children using 6,093 PIC records. Hu et al. (2026) used SHAP-interpreted ML on PIC data for mortality prediction in pediatric pneumonia. Bravo and Fajardo (2025) addressed class imbalance in general pediatric ICU mortality prediction on PIC using SHAP visualization.

A pattern is clearly evident: while the PIC database has been used with SHAP-based explainability across multiple clinical conditions, sepsis-specific onset prediction remains unaddressed. Furthermore, none of the above studies applied CatBoost as the primary classifier for sepsis. Song et al. (2024) used PIC data to analyze the prognostic value of early lactate measurement in pediatric sepsis — a clinical study rather than an ML prediction pipeline — further confirming that the modeling gap persists despite established clinical interest in this dataset for sepsis-related research.

4. Explainable Artificial Intelligence in Clinical Sepsis Prediction

The 'black-box' nature of high-performing ML models represents a fundamental barrier to clinical adoption. Clinicians require not only accurate predictions, but also transparent explanations of which features drive individual predictions — a prerequisite for trust, accountability, and integration into clinical decision support workflows. Explainable AI (XAI) methods, particularly SHapley Additive exPlanations (SHAP), have emerged as the dominant framework for post-hoc model interpretation in clinical ML (Athukorala & Ilmini, 2026).

SHAP decomposes each model prediction into individual feature contributions, allowing clinicians to identify which variables — and to what degree — influenced a given risk score. In adult sepsis contexts, SHAP analyses have consistently identified lactate, white blood cell count, serum albumin, and organ dysfunction markers as dominant predictors. Shi et al. (2025) applied SHAP to their XGBoost-based pediatric sepsis model and identified lactate, pH, WBC count, and vasopressor use as top predictors — findings concordant with established clinical understanding of sepsis pathophysiology and reinforcing the potential of SHAP to generate clinically actionable insights.

A 2026 systematic review of interpretable models for sepsis and ICU mortality prediction by Athukorala and Ilmini evaluated studies from 2020 to 2025 and found that while XAI methods maintain high predictive accuracy, their application remains disproportionately concentrated in adult Western ICU populations. Pediatric applications were identified as a priority research area, with the review explicitly recommending pediatric-specific XAI model development as a direction for future work. This systematic recommendation from a 2026 review directly supports the clinical and scientific rationale for the present study.

The human-in-the-loop framework proposed by Nagarajan et al. (2025) for pediatric sepsis prediction further underscores the importance of interpretability — arguing that ML models for pediatric sepsis must integrate clinical user perspectives and transparent feature attribution to

achieve real-world deployment. This framework explicitly calls for models that can communicate prediction rationale to bedside clinicians, a need directly addressed by SHAP-based pipelines.

5. Identified Research Gap and Proposed Contribution

A systematic review of the existing literature reveals a three-dimensional gap at the intersection of algorithm, dataset, and clinical task. First, while CatBoost has demonstrated strong performance for pediatric sepsis prediction, it has only been evaluated on private Western datasets (Chanci et al., 2025; Han et al., 2024), and never on the publicly available PIC database. Second, the PIC database has been used with SHAP-based explainability for AKI, DIC, pneumonia, and general mortality — but not for sepsis onset prediction. Third, ML studies using PIC data for sepsis have focused on survival time and mortality outcomes (Huang et al., 2025; Song et al., 2024) rather than early onset detection — a clinically distinct and arguably more actionable target.

Google Scholar searches confirmed the sparsity of this intersection: the query "PIC database" "sepsis" "CatBoost" returned only 4 results globally, none of which addressed sepsis onset prediction as a primary outcome. The query "pediatric sepsis" "PIC database" "machine learning" returned only 6 results, none combining CatBoost with SHAP for onset prediction. Independent synthesis by Elicit AI explicitly confirmed that 'the evidence base for explainable ML in pediatric sepsis remains sparse in Chinese ICU contexts, with most explainable sepsis research conducted in Western healthcare systems.'

This study therefore proposes the development of an explainable CatBoost model for early prediction of pediatric sepsis onset using the PIC database, with SHAP applied to identify clinically meaningful feature importance. The proposed approach offers three contributions: (1) it fills the unoccupied intersection of CatBoost, SHAP, PIC database, and pediatric sepsis onset prediction; (2) it provides a fully reproducible pipeline built on a publicly available pediatric Chinese ICU dataset, enabling external validation and benchmarking; and (3) it generates SHAP-derived clinical insights that are directly interpretable by PICU clinicians, supporting real-world decision support integration consistent with the framework advocated by Nagarajan et al. (2025) and the recommendations of Athukorala and Ilmini (2026).

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